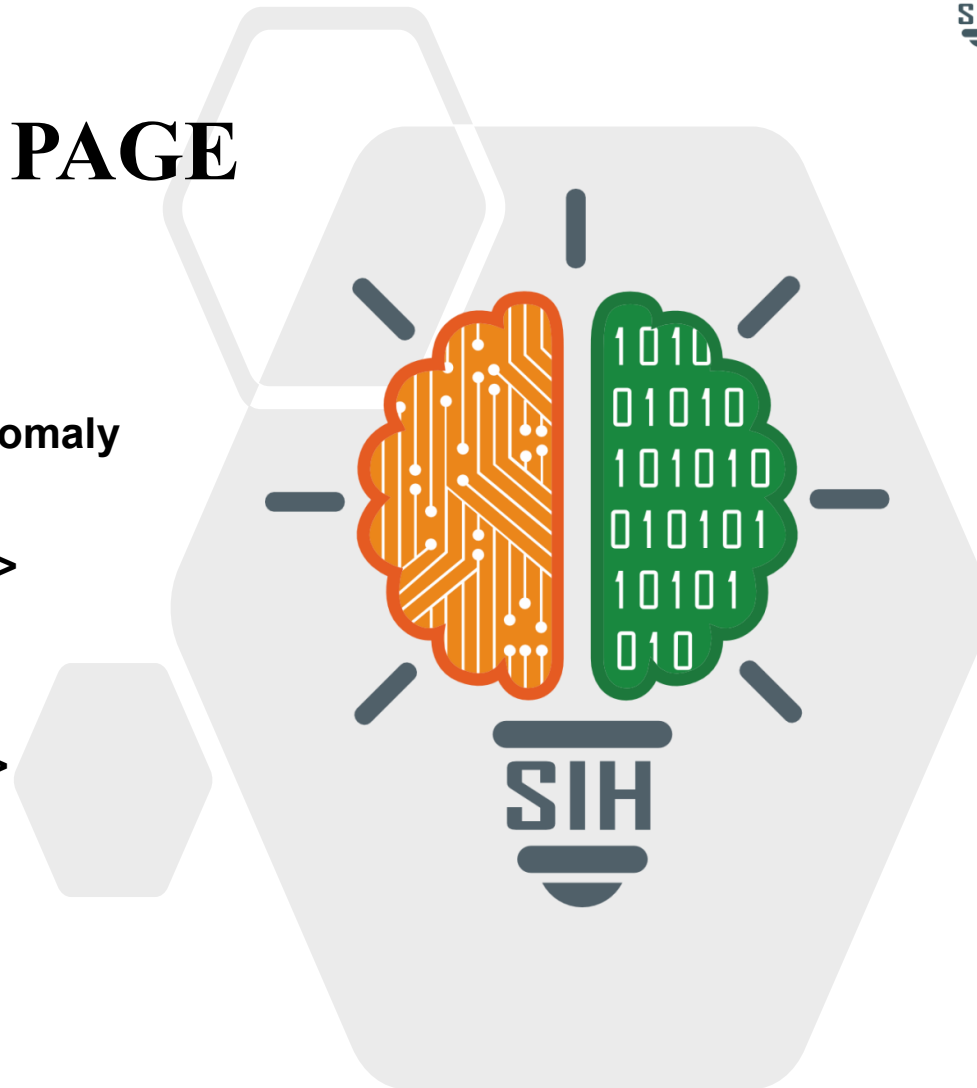


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – PS26073
- Problem Statement Title– AI/ML based anomaly detection in Automatic Weather Station data
- Theme– <THEME AS LISTED ON THE SIH PORTAL>
- PS Category- Software
- Team ID– <TEAM ID>
- Team Name (Registered on portal)– <TEAM NAME>



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

• Detailed explanation of the proposed solution

- Four stages: remove the expected daily and seasonal cycle, compare the station against its nearest neighbours, score what is left with an unsupervised model, then name the likely fault from its shape.
- Uses only the three fields the problem statement provides — temperature, pressure and relative humidity. No extra instrument is assumed.
- Two tiers: a physical-plausibility screen on the station node itself, and everything that needs neighbouring stations on one server.

• How it addresses the problem

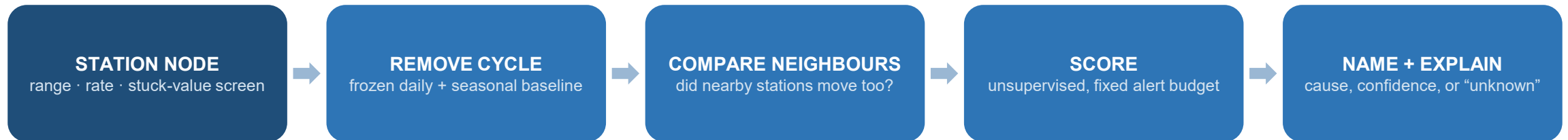
- Separates a faulty sensor from real weather by asking whether the neighbouring stations moved in the same direction at the same time.
- Every alert names a probable cause with a confidence — stuck probe, drift, step offset, dropout — instead of raising a bare flag.
- Alert volume is held to a fixed budget per station per week, so the output stays small enough for an operator to actually read.

• Innovation and uniqueness of the solution

- The baseline is frozen on an approved reference window and never refitted on live data: a self-updating baseline quietly absorbs slow drift and hides the very fault it should expose.
- Dew point is used as a physical cross-check — heating a probe raises temperature and lowers humidity but leaves dew point where it was, which separates a shield or siting fault from genuinely warmer air.
- Answering “unknown” when the evidence supports no cause is a designed output, not a failure.

TECHNICAL APPROACH

- **Technologies to be used (e.g. programming languages, frameworks, hardware)**
 - Python — NumPy, pandas and scikit-learn for the residual, detection and classification stages. No GPU and no model server.
 - FastAPI service with a browser operator console; Leaflet map on satellite imagery with the official India boundary served by NCMRWF (MoES).
 - Station node: ESP32-class microcontroller with a BME280 temperature / pressure / humidity sensor. PostgreSQL with TimescaleDB for storage.
- **Methodology and process for implementation (Flow Charts/Images/ working prototype)**
 - The station tier holds only constant-size state, so no history buffer or model runtime is needed on the node; thresholds are calibrated on a clean reference period before deployment.



FEASIBILITY AND VIABILITY



- **Analysis of the feasibility of the idea**

- A national network at 15-minute reporting is only a few records per second — one modest on-premise VM handles it with no GPU.
- Methods are established and inspectable: robust regression, neighbour differencing, unsupervised scoring. Not a black box.
- The station tier is inexpensive and runs on hardware already common in AWS deployments.

- **Potential challenges and risks**

- Station density — neighbour comparison only works where a nearby station is correlated enough to serve as a reference.
- Genuine severe weather can look like a fault, risking false alarms during exactly the events that matter most.
- Slow calibration drift never looks like an outlier at any single moment.
- Alert fatigue: a system that flags too much gets switched off.

- **Strategies**

- Measure how inter-station differences grow with distance first, and let that measurement choose each station's neighbours.
- Gate alerts on neighbour agreement, so a change the whole region shares reads as weather rather than a fault.
- Freeze the baseline on approved data and track drift as a divergence growing over days, never from a single reading.
- Cap alerts to a fixed budget per station-week, and abstain when uncertain.

IMPACT AND BENEFITS



- **Potential impact on the target audience**
 - IMD and MoES network operators get a ranked, named list of the stations needing attention, instead of raw observation streams to inspect.
 - Forecasting and numerical weather prediction gain cleaner input: bad observations are caught before they are assimilated.
 - Field maintenance is directed to the station that is actually faulty, and to the sensor within it.
- **Benefits of the solution (social, economic, environmental, etc.)**
 - Social — warnings and forecasts that reach the public rest on observations checked for sensor faults.
 - Economic — fewer wasted site visits, and instrument replacement planned from evidence of degradation rather than a fixed cycle.
 - Environmental — the node transmits only flagged readings and a periodic health summary; radio transmission dominates an AWS power budget, so a quieter uplink means a smaller power system.
 - Institutional — a long-term record whose quality is documented, which matters for climate-scale reuse of the archive.

RESEARCH AND REFERENCES



- **Details / Links of the reference and research work**

- IMD — M. N. Ranalkar et al., “Quality control of Automatic Weather Station data”, MAUSAM 66(1), 2015: the checks IMD applies today, including the spatial regression test.
- IMD Vision 2047 (press release, 14 Jan 2025) — smart weather stations, AI for real-time quality control, and self-diagnosing instruments: internal.imd.gov.in/press_release/20250114_pr_3552.pdf
- NCMRWF Technical Report NMRF/TR/02/2022 — observation quality control and buddy checks in the operational assimilation system.
- NCMRWF operational dashboard and model guidance: nwp.ncmrwf.gov.in
- WMO-No. 8, Guide to Instruments and Methods of Observation (CIMO Guide) — measurement uncertainty and siting classification.
- ECMWF ERA5 reanalysis via the Copernicus Climate Data Store — reference fields used to develop and test the method.